

Product Requirements Document: Construction Payroll System (Tier 4)

1. Introduction

This document outlines the functional requirements for a Tier 4 construction payroll system, designed to manage Very High Complexity scenarios. This system is for contractors who employ union labor on government-funded projects, making them subject to both Collective Bargaining Agreements (CBAs) and prevailing wage laws simultaneously.¹ This is the most challenging payroll environment, requiring the system to integrate, compare, and correctly apply two distinct and complex sets of rules. This PRD assumes all features from Tiers 1, 2, and 3 are present and serves as the final layer of required functionality.

2. Target Users

The target users remain consistent with previous tiers, but their need for precision, compliance assurance, and advanced reporting is at its peak:

- **Payroll Administrators & Compliance Managers:** Primary users who will configure and manage the complex logic for comparing union and prevailing wage rates, generate dual compliance reports, and conduct internal audits.
- **Company Owners/Executives:** Require absolute assurance of compliance to mitigate the significant financial and legal risks associated with errors in this tier.
- **Project Managers:** Need fully burdened labor cost reports that accurately reflect the "higher of" calculations to manage budgets on these high-stakes projects.

3. Functional Requirements

The following requirements are additive. All features from Tier 1 (Core), Tier 2 (Union), and Tier 3 (Certified) are prerequisites for this system.

3.1. Prerequisite Functionality

- All functional requirements from the **Tier 1 (Core Payroll)** system are required.
- All functional requirements from the **Tier 2 (Non-Certified, Union Payroll)** system are required.
- All functional requirements from the **Tier 3 (Certified, Non-Union Payroll)** system are required.

3.2. Integrated Prevailing Wage & Union Compliance

This module contains the advanced logic engine required to navigate the overlapping requirements of CBAs and prevailing wage laws.¹

- **FR-VH1.1: Simultaneous Rule Application:** The system must be able to apply both union

CBA rules and prevailing wage determinations to a single employee on a single project for the same pay period.¹

- **FR-VH1.2: Advanced "Higher Of" Logic:** The system must feature an automated comparison engine that, for each employee and each hour worked, calculates and applies the higher of the two required rates for both wages and benefits.¹ This includes comparing:
 - The union base wage rate vs. the prevailing wage base rate.
 - The total value of union fringe contributions vs. the prevailing wage fringe rate.
- **FR-VH1.3: Automated "Top-Up" Calculation:** If the total value of the union compensation package (wage + fringes) is less than the total required by the prevailing wage determination, the system must automatically calculate the shortfall and pay the difference ("top-up") to the employee.¹ The system must support paying this difference as additional cash wages or, if allowed, by adjusting contributions to other bona fide benefit plans.
- **FR-VH1.4: Integrated Apprenticeship Management:** The system must be able to manage union apprenticeship programs in conjunction with prevailing wage apprentice requirements, correctly applying the appropriate ratios and rates from both sets of rules.¹

3.3. Enhanced Certified Payroll Reporting (Union Context)

This module ensures that the mandatory certified payroll reports accurately reflect the complex, dual-compliance calculations.¹

- **FR-VH2.1: Comprehensive Certified Reports:** The system must generate Federal WH-347 and state-equivalent certified payroll reports that accurately display¹:
 - Both the union classification and the prevailing wage classification for the worker.
 - The final hourly rate paid, which must include any cash top-ups required to meet the prevailing wage.
 - A detailed breakdown of all fringe benefits, clearly distinguishing between union-contributed benefits and any additional cash-in-lieu payments.
- **FR-VH2.2: Automated Rate Verification:** The system must include an automated check to verify that the final reported "straight time" and "overtime" rates meet the requirements of *both* the CBA and the prevailing wage determination.¹

3.4. Advanced Reporting & Auditing

This module provides the tools necessary to audit and verify compliance across both union and government requirements.¹

- **FR-VH3.1: Cross-Referencing Reports:** The system must be able to generate reports that cross-reference data between the Union Remittance Reports and the Certified Payroll Reports to ensure complete consistency and accuracy.¹
- **FR-VH3.2: Detailed Audit Trails:** Must provide detailed, easily accessible audit trails for all wage calculations, especially the "higher of" rate comparisons and any top-up payments, to support internal or external audits.¹
- **FR-VH3.3: Proactive Discrepancy Reporting:** The system must offer comprehensive

discrepancy reports that can be run before payroll is finalized, highlighting any potential non-compliance issues between the two sets of rules.¹

4. Non-Functional Requirements

- **NFR-P1: Performance:** The system must process payroll for 500 employees, including complex "higher of" logic, within 45 minutes.
- **NFR-P2: Reporting Performance:** Must generate a complex, dual-compliance certified report (e.g., WH-347 for 200 union employees) in under 5 minutes.
- **NFR-S3: Security:** Must feature role-based access control (RBAC) to ensure data confidentiality. All sensitive employee, union, and project data must be encrypted at rest and in transit.
- **NFR-U1: Usability:** The interface for configuring the "higher of" logic and reviewing discrepancy reports must be intuitive to minimize the risk of human error in these high-stakes calculations.
- **NFR-R1: Reliability:** Must maintain 99.9% uptime for all core payroll and reporting features.
- **NFR-I1: Integration:** Must offer a robust API for integration with common construction accounting/ERP systems to ensure that the final, fully burdened labor costs are accurately reflected in job cost reports.